

INVESTIGATION REPORT

Issue 1011: Debug entry with no recorded cause

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1011
Disposition	Dead code, proven
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1011
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL master 6033d2b, COREV_PULP=1 FPU=0
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	6 cited, 6 present

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

Both 'all-false' else-branches are unreachable in the default CV32E40P configuration (COREV_PULP=0, COREV_CLUSTER=0, FPU=0). Line 1187: state DBG_TAKEN_ID is entered only from DBG_FLUSH when debug_mode_q, trigger_match_i, (ebrk_force_debug_mode & ebrk_insn_i), data_load_event_i (constant 0 without COREV_CLUSTER) or the sticky debug_req_entry_q holds; with ~debug_mode_q one of the three causes is therefore always still true one cycle later, so the if/else-if chain always takes a branch. Line 1210: every transition into DBG_TAKEN_IF either sets debug_force_wakeup_n=1 (BOOT_SET, SLEEP, wfi) or is gated by debug_single_step_i, and no reachable state clears dcsr.step in between. An unbounded PDR proof (abc pdr under yosys/sby) of each negated condition passes in 25.4 s and 17.6 s, and bounded covers show both enclosing states are reachable (step 8 and step 5), so the proofs are not vacuous. Assumptions: OBI outstanding-request handshake, scan_cg_en_i=0, fetch_enable_i=1 (flows/constraints_cv32e40p.sv). Recommend waiving both holes as unreachable (matches the issue's own OneSpin conclusion in spirit; this run did not see that evidence).

3 Conditions

- 1. OBI outstanding-request handshake
- 2. scan_cg_en_i=0
- 3. fetch_enable_i=1 (flows/constraints_cv32e40p

Extracted verbatim from the environment assumptions stated in the verdict record for this run.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:1187-1191 (DBG_TAKEN_ID) and :1210-1212 (DBG_TAKEN_IF): the all-false branch of each if/else-if chain is unreachable. DBG_TAKEN_ID is only entered from DBG_FLUSH when one of trigger_match_i / (ebrk_force_debug_mode & ebrk_insn_i) / debug_req_entry_q (or debug_mode_q, which skips the chain) holds, and ID is halted so the cause is still present in DBG_TAKEN_ID. DBG_TAKEN_IF is only entered with debug_force_wakeup_n=1 (BOOT_SET/SLEEP/wfi) or under debug_single_step_i, so one branch is always taken. Proven unbounded with open-source formal (yosys+sby, abc pdr) in the default configuration; anti-vacuity covers of both states reached. Waive as unreachable code (branch coverage).

Checker used

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runs/v1-cv32e40p-1011/checker_1011.sv
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6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	runs/v1-cv32e40p-1011/formal/p_assert_1187/sby.png	present	abc pdr PASS (unbounded): DBG_TAKEN_ID && !debug_mode_q never coexists with all of trigger_match_i, ebrk_force_debug_mode&ebrk_insn_i, debug_req_entry_q false. 25.4 s.
E2	proof	runs/v1-cv32e40p-1011/formal/p_assert_1210/sby.png	present	abc pdr PASS (unbounded): DBG_TAKEN_IF never coexists with debug_force_wakeup_q=0 and debug_single_step_i=0. 17.6 s.
E3	proof	runs/v1-cv32e40p-1011/formal/c_state_1187/sby.png	present	Anti-vacuity: covers DBG_TAKEN_ID and DBG_TAKEN_ID&&!debug_mode_q both reached at step 8 (trace job/engine_0/trace0.vcd, rendered cover_dbg_taken_id.png).
E4	proof	runs/v1-cv32e40p-1011/formal/c_state_1210/sby.png	present	Anti-vacuity: cover DBG_TAKEN_IF reached at step 5 (trace job/engine_0/trace0.vcd, rendered cover_dbg_taken_if.png).
E5	waveform-diff	runs/v1-cv32e40p-1011/cover_dbg_taken_id.png	present	Cover trace: debug_req_i asserted, debug_req_entry_q set, FSM reaches DBG_TAKEN_ID with debug_mode_q=0.
E6	waveform-diff	runs/v1-cv32e40p-1011/cover_dbg_taken_if.png	present	Cover trace: debug_req_i during boot -> BOOT_SET -> DBG_TAKEN_IF with debug_force_wakeup_q=1.

Paths are relative to runs/v1-cv32e40p-1011. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

This run records evidence paths repository-root-relative rather than relative to its own run directory. They were resolved on that basis. The inconsistency is in the recorded data, not in the evidence.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- waveform-diff. Signal-level comparison between two runs.

7 Limitations and scope

- The proof establishes unreachable under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.

- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

Run directory	runs/v1-cv32e40p-1011
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1011.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024: waived; formal assert all_true_ctrl_1187_1189_and_1191 / 1210_and_1212 in scripts/formal
Assessed confidence	0.9. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	1

Generated by reporting/scripts/issue_document.py from runs/v1-cv32e40p-1011/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.